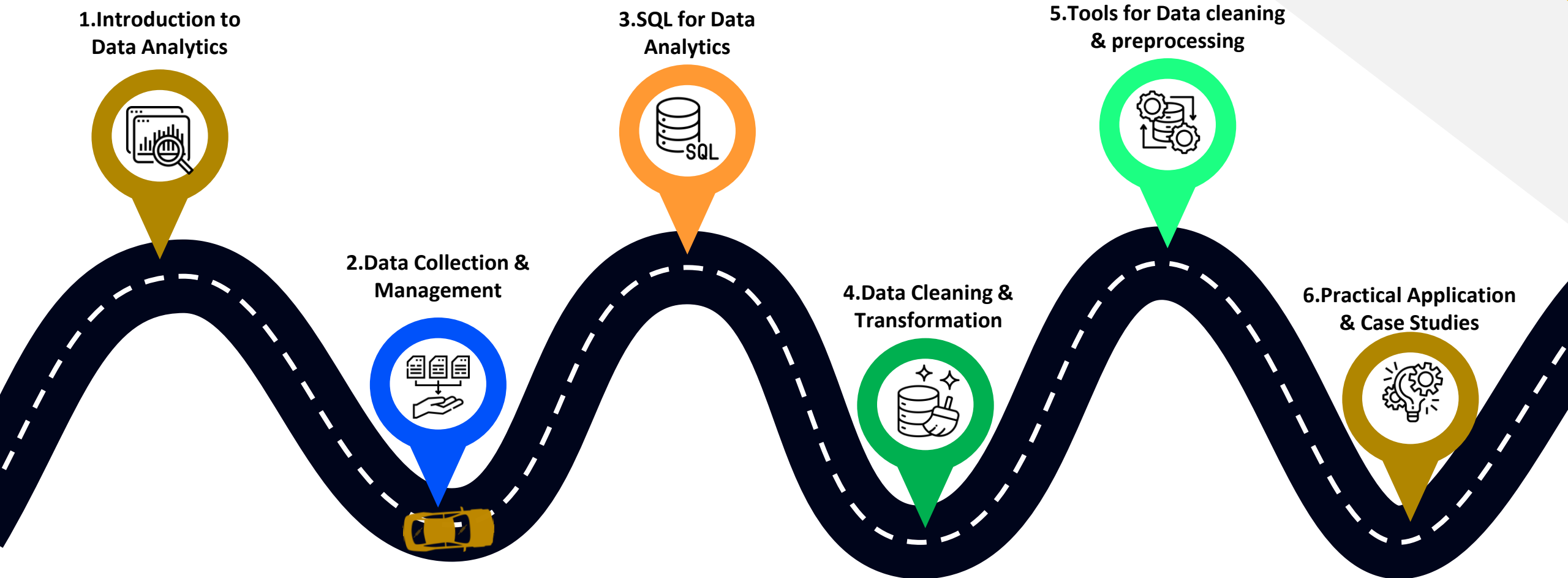




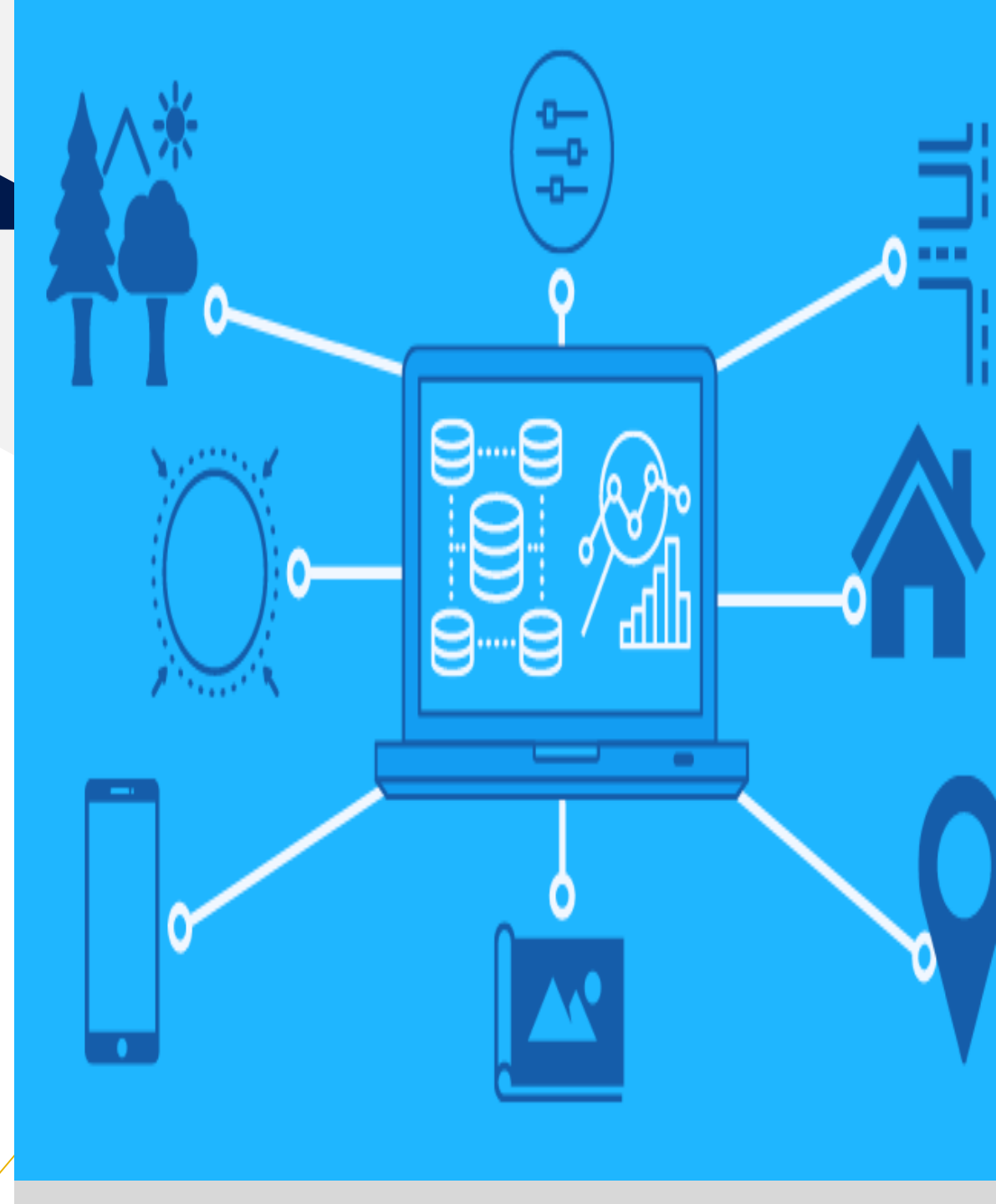
## **Part Two: Data Collection and Management**

# Course Road Map



# Data Sources

- Internal VS External Sources
- First-Party, Second-Party and Third-Party Data
- Real Time VS Batch Data



# Internal Data Sources

- Company databases, CRM systems, ERP systems.
- Benefits and challenges.





# External Data Sources

- Social media, public datasets, market research.
- Benefits and challenges.



# First-Party, Second-Party and Third-Party Data

| Aspect     | First-Party Data                                                      | Second-Party Data                                                 | Third-Party Data                                                               |
|------------|-----------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Definition | Data collected directly from your customers.                          | Another business's first-party data shared with you.              | Data collected by an external entity and sold to businesses.                   |
| Examples   | Customer names, emails, purchase history.                             | Data from a partner with a similar audience.                      | Demographic data from a broad population.                                      |
| Benefits   | Highly accurate and relevant, provides detailed customer insights.    | Access to new customer segments, insights from similar audiences. | Large datasets, helps in broad targeting and expanding reach.                  |
| Drawbacks  | Limited to existing customers, not useful for reaching new audiences. | Potential privacy issues, data quality depends on the partner.    | Variable accuracy, potential privacy concerns, less specific to your audience. |

# Real Time VS Batch Data

| Aspect     | Batch Data                                                            | Real-Time Data                                             |
|------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| Definition | Data collected and processed in large groups at scheduled intervals.  | Data collected and processed continuously as events occur. |
| Examples   | Daily sales reports, monthly financial statements.                    | Live stock market feeds, real-time social media analytics. |
| Benefits   | Efficient for large volumes of data, lower processing cost.           | Immediate insights, suitable for time-sensitive decisions. |
| Drawbacks  | Delay in data availability, not suitable for time-sensitive insights. | Higher processing cost, more complex infrastructure.       |

# Types Of Data Structure

- **Structured Data**
- **Unstructured Data**
- **Semi-Structured Data**





# Types Of Data Structure

| Aspect                     | Structured Data                                                         | Semi-structured Data                                                 | Unstructured Data                                                                   |
|----------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Definition</b>          | Data organized into rows and columns, easily searchable and analyzable. | Data that has some organizational properties but not a fixed schema. | Data that lacks a predefined format or structure.                                   |
| <b>Examples</b>            | Relational databases, spreadsheets.                                     | JSON files, XML files, email.                                        | Text documents, images, videos, social media posts.                                 |
| <b>Tools</b>               | SQL databases (MySQL, PostgreSQL), Excel.                               | NoSQL databases (MongoDB, Cassandra), data interchange formats.      | Big data tools (Hadoop), text analytics tools, multimedia analysis tools.           |
| <b>Real-World Use Case</b> | Financial transactions, inventory management.                           | Log files, configuration files, data integration tasks.              | Customer feedback analysis, multimedia content management, social media monitoring. |

# Types Of Data Structure

```

1 <order>
2   <order_id>1024</order_id>
3 <customer>
4   <first_name>Sara</first_name>
5   <last_name>Tarek</last_name>
6 </customer>
7 <items>
8 <item>
9   <product>Cappuccino</product>
10  <quantity>2</quantity>
11  <price>40</price>
12 </item>
13 <item>
14  <product>Croissant</product>
15  <quantity>1</quantity>
16  <price>20</price>
17 </item>
18 </items>
19 <total>100</total>
20 <promo_code_used>true</promo_code_used>
21 </order>

```

**XML**

```

{
  "order_id": 1024,
  "customer": {
    "first_name": "Sara",
    "last_name": "Tarek"
  },
  "items": [
    {
      "product": "Cappuccino",
      "quantity": 2,
      "price": 40
    },
    {
      "product": "Croissant",
      "quantity": 1,
      "price": 20
    }
  ],
  "total": 100,
  "promo_code_used": true
}

```

**JSON**

# Questions

| Data Example                                          | Source | Processing | Structure |
|-------------------------------------------------------|--------|------------|-----------|
| Customer purchase records from a company's ERP system | ?      | ?          |           |
| Weather data from an API like OpenWeatherMap          | ?      | ?          |           |
| Comments from your company's Instagram post           | ?      | ?          |           |
| A survey response form in Excel                       | ?      | ?          |           |
| Emails between the sales team and clients             | ?      | ?          |           |
| JSON file from a public COVID-19 API                  | ?      | ?          |           |
| Video recordings of customer service calls            | ?      | ?          |           |

# Databases, APIs, and web scraping

- **Database Key Concepts**
- **Relational Database**
- **Non-Relational Database**
- **APIs & Web Scraping**





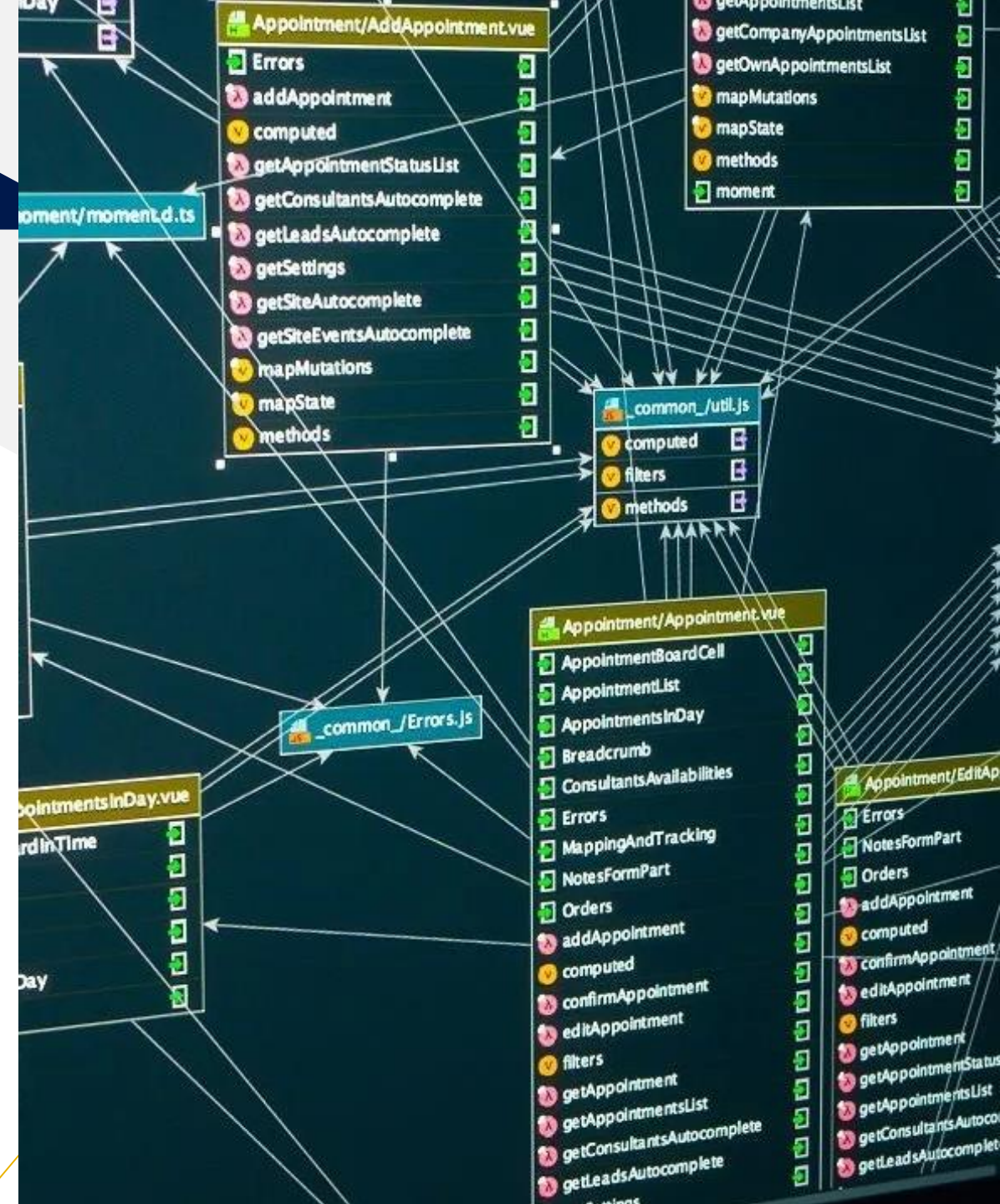
# Database Key Concepts

- Database Definition and Main Components
- DBMS and Data Models



# Database Definition and Main Components

- A **Database** is an organized collection of data, typically stored and accessed electronically from a computer system. Databases are designed to manage, store, and retrieve large amounts of information efficiently, ensuring data integrity, consistency, and security.





# DBMS and Data Models

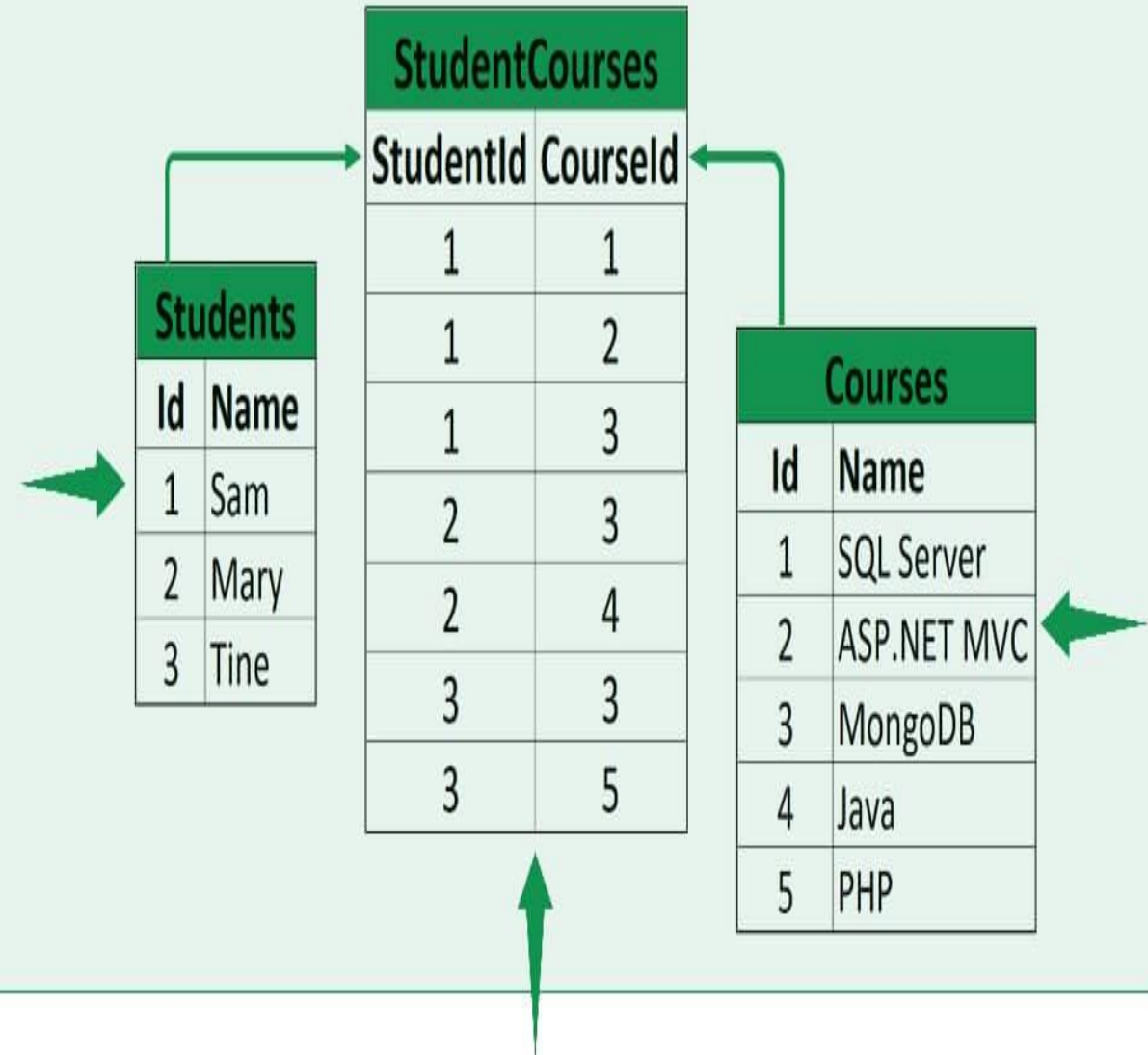
- A **Database Management System (DBMS)** is software that interacts with end-users, applications, and the database itself to capture and analyze data.
- The DBMS provides a systematic way to create, retrieve, update, and manage data. It ensures data integrity, consistency, security, and efficient data management.
- Examples: MySQL, PostgreSQL, Microsoft SQL Server, and MongoDB



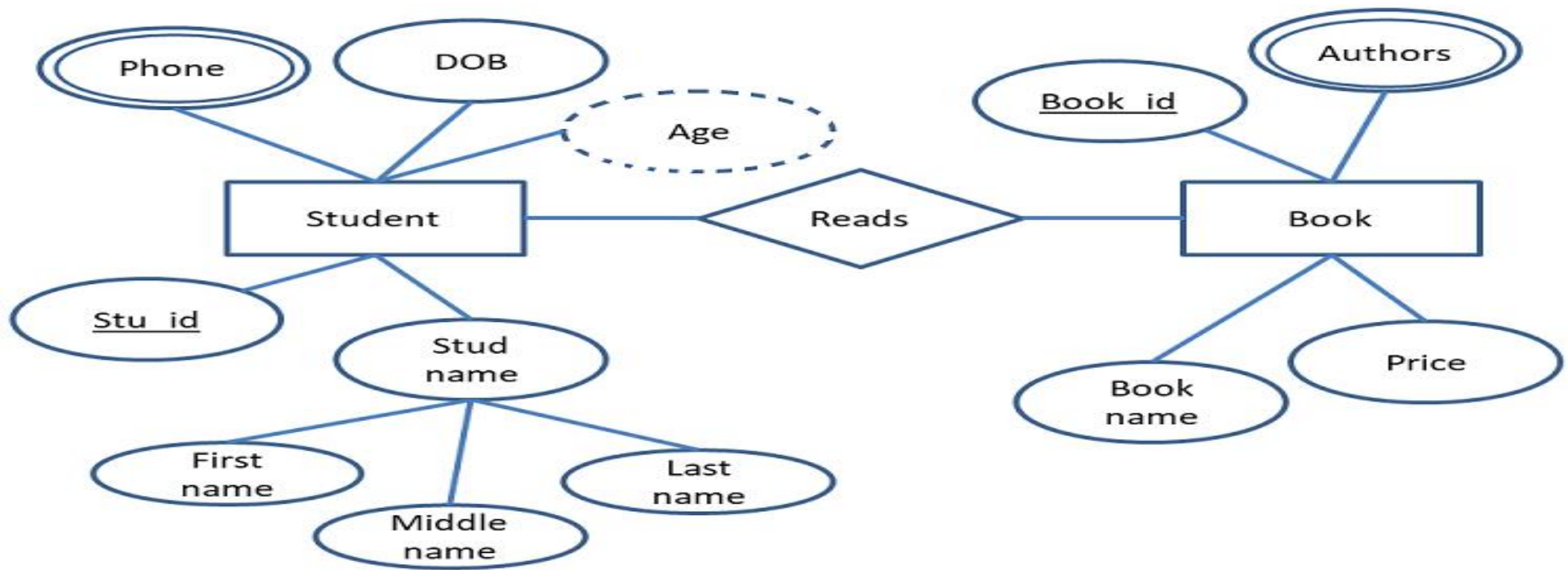
# Relational Database

- Entity Relationship Diagram (ERD)
- Relationships Degree
- Cardinality Ratio
- Normalization

## Relational Database

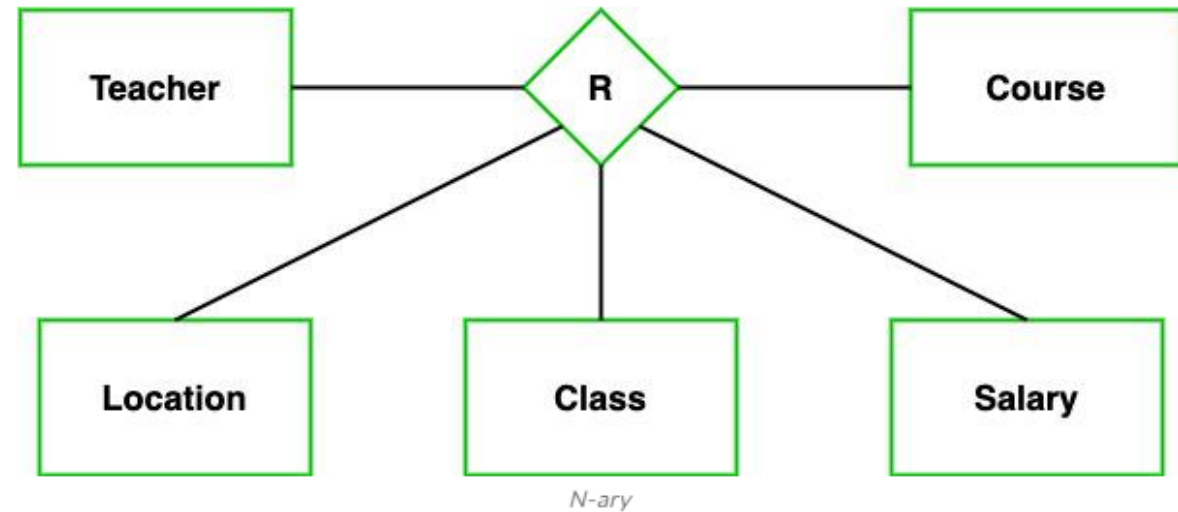
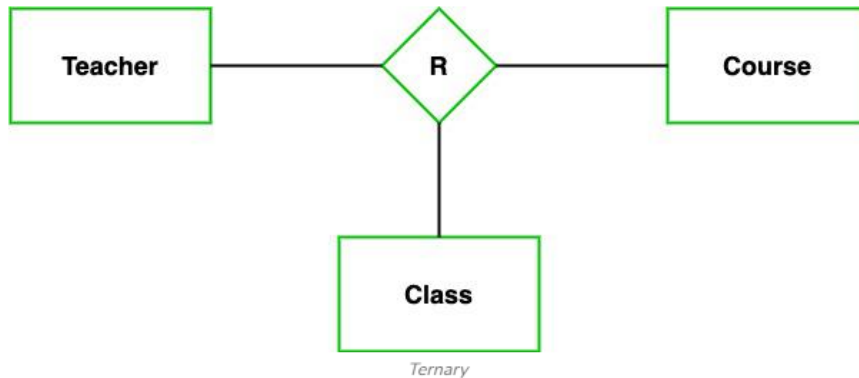
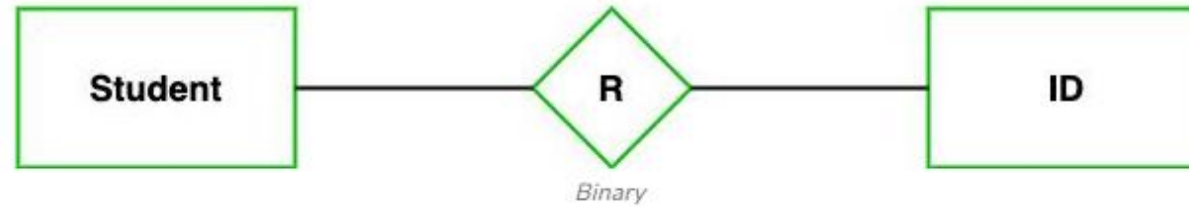
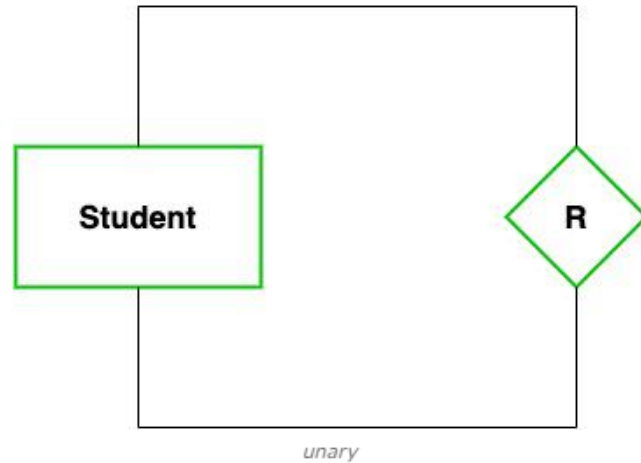


# Entity Relationship Diagram (ERD)





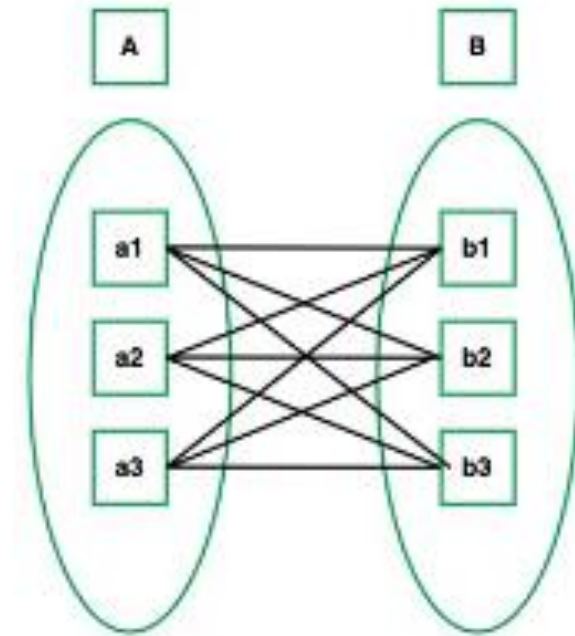
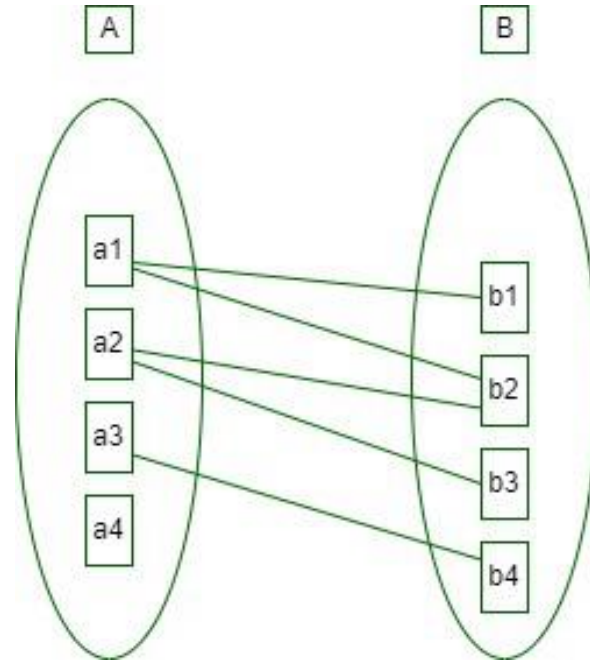
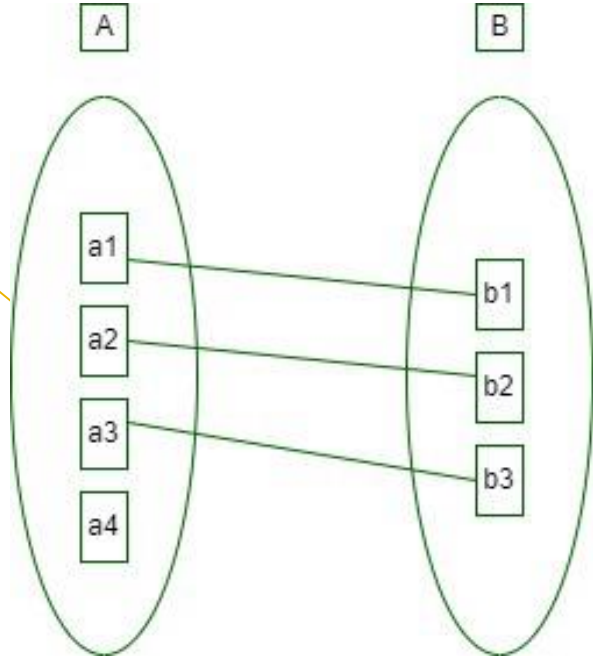
# Relationships Degree



# Relationships Degree

| Aspect     | Unary (Degree 1)                         | Binary (Degree 2)                     | Ternary (Degree 3)                             | N-ary (n Degree)                                      |
|------------|------------------------------------------|---------------------------------------|------------------------------------------------|-------------------------------------------------------|
| Definition | Relationship within a single entity type | Relationship between two entity types | Relationship involving three entity types      | Relationship involving more than three entity types   |
| Examples   | Employee manages Employee                | Customer places Order                 | Supplier supplies Product to Warehouse         | Company finances Project with Funding from Investors  |
| Use Cases  | Self-referencing relationships           | Common in relational databases        | Complex relationships involving three entities | Advanced database schemas involving multiple entities |

# Cardinality Ratio



# Cardinality Ratio

- One to One Relationship
- One to Many Relationship
- Many to many Relationship





# Case Study

